

Bridging research and policy in China's energy sector: A semantic and reinforcement learning framework

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ABSTRACT

Aligning academic research with policymaking is vital for addressing China's energy challenges. This study introduces an AI-driven framework combining BERTopic, semantic similarity analysis, and deep reinforcement learning (DRL) to evaluate alignment between 106,661 English-language academic papers and 618 national-level policy documents. Topic modeling reveals strong convergence in themes such as "Coal mining and geological formations," which account for 15.73 % of academic publications, while "Safety regulations and worker protection" dominate policy texts at 13.35 %. In contrast, emerging topics like "Digital economy and carbon transformation" remain underrepresented, with a popularity score of only 0.13. Semantic similarity analysis across 22 policy and 27 academic topics yields an average cosine similarity of 0.23, with only 12.5 % of topic pairs exceeding 0.4, underscoring thematic misalignment. Structurally, policy networks are 15.9 times denser and exhibit $30 \times$ higher clustering coefficients than scientific networks, indicating more centralized but less diversified discourse. DRL-based prioritization identifies "Power systems and renewable integration" as the top-performing theme (Q-value = 1.6225), highlighting opportunities for targeted energy transition policies. These quantitative results offer empirical evidence to guide theme-based policy adaptation and foster actionable science-policy integration.

1. Introduction

The global energy landscape faces unprecedented challenges, including ensuring sustainability, enhancing energy security, and transitioning to renewable energy [1,2]. As a major energy consumer and emitter of greenhouse gases, China plays a pivotal role in shaping the global energy transition [3]. However, the persistent disconnect between academic research and policymaking in the energy sector has hindered the effective translation of innovative findings into impactful policies [4]. While academic studies often emphasize long-term, exploratory solutions, policymakers prioritize actionable frameworks that address immediate challenges [5–7]. Bridging this gap is essential for tackling the pressing issues of energy governance and sustainable development.

Existing studies have provided valuable insights into thematic trends and policy alignment through bibliometric and qualitative approaches [8]. However, traditional methods frequently treat policies as static entities, failing to capture their dynamic evolution or identify emerging thematic trends [9,10]. These limitations impede their capacity to provide actionable guidance in rapidly evolving contexts. Moreover, while

some studies address thematic alignment, they often lack granularity and overlook structural dynamics between academic and policy networks [11]. This challenge is particularly salient in the context of China, where both academic output and national-level policy interventions in the energy sector have grown exponentially [12]. As one of the world's largest energy consumer and carbon emitter [13], China plays a central role in the global energy transition [14], yet faces persistent disconnects between scientific research and policy implementation. Despite this surge in activity, there remains a lack of effective, scalable tools to assess the semantic and structural alignment between academic insights and evolving policy priorities. Existing approaches tend to be static, fragmented, or descriptive in nature, limiting their ability to inform dynamic decision-making [15]. This study is thus motivated to develop an integrated, AI-enhanced framework that leverages recent advances in natural language processing and reinforcement learning to analyze thematic convergence, identify research-policy mismatches, and support real-time prioritization. Focusing on China's energy sector provides an ideal testing ground for this framework, given its centralized policy-making structure and data-rich environment [16].

Prior studies examining the misalignment between scientific

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research and policy-making in fields such as climate and energy have primarily employed bibliometric analysis, expert evaluation, and static topic modeling techniques like LDA [9,17]. While useful, these methods often lack semantic depth, temporal sensitivity, and interpretability—particularly in fast-evolving policy domains [18]. In response, this study adopts BERTopic, which integrates transformer-based sentence embeddings with class-based TF-IDF to generate context-aware and temporally dynamic topics, enabling finer-grained thematic tracking across research and policy corpora [19]. Additionally, a deep reinforcement learning (DRL) module is introduced to dynamically rank topics based on alignment trajectories and relevance, thus providing an adaptive mechanism for prioritizing themes over time. By combining these advances, the study aims to develop a scalable, AI-driven framework that not only maps thematic convergence in China's energy sector but also supports real-time policy decision-making—offering a clear departure from previous static and descriptive approaches.

This study makes three key contributions.

- It proposes a dynamic and interpretable framework to assess research-policy alignment in China's energy sector, combining semantic modeling with deep reinforcement learning for real-time topic prioritization.
- It constructs a curated dataset of English-language academic publications and centralized national-level policy documents, enabling large-scale empirical analysis.
- It integrates natural language processing, bibliometrics, statistical methods, and machine learning into a unified pipeline, offering a novel interdisciplinary approach to energy policy analytics.

Beyond its application to China's energy policy, the proposed framework is adaptable to other domains such as environmental governance and public health, where aligning research and policy is equally critical. This study lays a foundation for fostering interdisciplinary collaboration and driving innovation, offering policymakers a robust tool to navigate the complexities of contemporary energy governance.

The remainder of this article is structured as follows. The second section presents the methods and materials, including data sources, topic modeling, and the AI-based alignment framework. The third section reports the empirical results of the alignment analysis and dynamic prioritization. The fourth section discusses the key findings and theoretical implications. The fifth and final section concludes the study and

outlines its limitations and directions for future research.

2. Materials and methods

This study employs a comprehensive data-driven framework to evaluate the alignment between academic research and policymaking in China's energy sector. The methodology integrates semantic and structural perspectives to capture both the thematic content and dynamic interactions within academic and policy networks. The framework comprises six stages: data collection, topic modeling, thematic alignment analysis, network coupling evaluation, dynamic decision-making using deep reinforcement learning (DRL), and actionable insights generation. The flowchart is shown in Fig. 1.

2.1. Data source

To ensure the reliability of the research materials, at the beginning, this study exclusively selected English-language articles and reviews from the Web of Science (WoS) platform, specifically utilizing the Science Citation Index Expanded (SCIE) and the Social Sciences Citation Index (SSCI) databases. These databases are esteemed for their rigorous peer-review processes [20,21]. The literature search strategy encompassed topic searches within titles, abstracts, Keywords Plus, and author keywords. Inclusion criteria were restricted to English-language articles of the type "article," spanning from the earliest relevant publications in the field up to December 31, 2024. The search keywords were aligned with policy documents found on Chinese government websites, including: "China Coal" (20,907 records), "China Electricity" (9700 records), "China energy" (63,821 records), "China Energy Conservation" (3674 records), "China Renewable Energy" (9050 records), and "China Petroleum and Natural Gas" (804 records). Following an initial aggregation, records lacking abstracts were excluded, resulting in a final dataset of 106,661 articles.

Policy data were collected from the National Energy Administration of China's official website (<https://www.nea.gov.cn/>), focusing on legislative and regulatory documents. The same keywords were used to align with the academic dataset. The dataset captures national-level policies but excludes regional and industry-specific regulations, which may limit comprehensiveness. While the selected sources ensure reliability, the exclusive reliance on English-language academic articles and centralized policy repositories introduces potential biases. Future research should include multilingual data and regional policies to enhance generalizability.

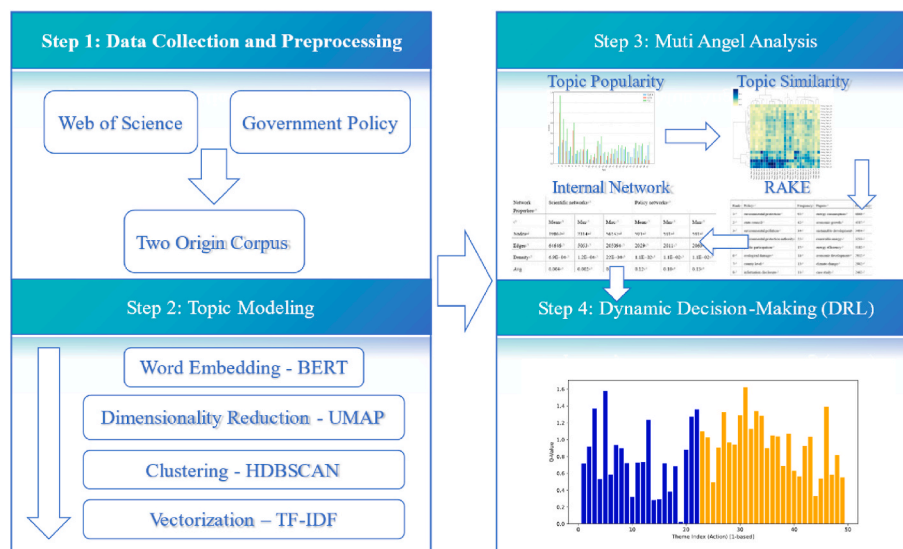


Fig. 1. Methodological flowchart of the study.

Additionally, the policy documents retrieved from the official repository represent only the most recent versions available as of 2024. Historical archives or versioned records are not publicly accessible, which restricts the ability to perform a longitudinal analysis of policy evolution. Consequently, this study is limited to a cross-sectional evaluation of current policy priorities. Future efforts may seek to construct synthetic policy timelines or leverage alternative governmental databases to enable time-series analyses of policy changes.

2.2. Topic finding

To identify thematic trends, BERTopic was employed for topic modeling [19]. BERTopic integrates natural language processing techniques with clustering algorithms to extract coherent and interpretable topics from textual data. The process begins with the application of Sentence-Transformer models to generate dense semantic embeddings for each document, ensuring a robust and high-quality representation of the textual content [22]. Following this, UMAP (Uniform Manifold Approximation and Projection) is employed for dimensionality reduction, which facilitates efficient clustering while preserving the inherent structure of the data [23]. Subsequently, HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) is applied to detect clusters within the reduced-dimensional space, enabling the determination of the optimal number of topics and effectively managing outliers [24]. Finally, BERTopic conducts a vectorization step, utilizing techniques such as Term Frequency-Inverse Document Frequency (TF-IDF) to extract and rank the most representative keywords for each identified topic, thereby ensuring their interpretability.

2.3. Topic popularity

After determining the topic distributions for each corpus, this study utilizes comprehensive metrics in the subsequent phase to evaluate the popularity of each topic, following the approach of Xiong et al. [25].

$$P^k = S_{NP}^k + S_{NTr}^k \quad (1)$$

$$S_{NP}^k = (P_A^k - P_A^{min}) / (P_A^{max} - P_A^{min}) \quad (2)$$

$$S_{Tr}^k = \frac{\sum_{y=2014}^{2024} \theta_k^y}{\sum_{y=2004}^{2014} \theta_k^y} \quad (3)$$

$$S_{NTr}^k = (S_{Tr}^k - S_{Tr}^{min}) / (S_{Tr}^{max} - S_{Tr}^{min}) \quad (4)$$

The ultimate popularity score P^k for each topic k is calculated by combining two normalized components: S_{NP}^k and S_{NTr}^k (Equation (1)). The first component, S_{NP}^k (Equation (2)), normalizes the proportion P_A^k of topic k by scaling it between 0 and 1 using the minimum P_A^{min} and maximum P_A^{max} proportions observed. This normalization ensures that the proportion is comparable across different topics. The second component, S_{Tr}^k (Equation (3)), captures the temporal dynamics by comparing the cumulative value θ_k^y of topic k from 2014 to 2024 against the baseline period from 2004 to 2014. This trend score highlights changes in the topic's prominence over time. To make this trend comparable, S_{Tr}^k is further normalized to S_{NTr}^k (Equation (4)) using the minimum S_{Tr}^{min} and maximum S_{Tr}^{max} trend scores.

2.4. Thematic alignment and semantic similarity

Semantic alignment between academic and policy topics was analyzed at both the corpus and topic levels to evaluate thematic alignment and semantic similarity. At the corpus level, the mean embeddings of the academic and policy corpora were computed using the paraphrase-mpnet-base-v2 model [22]. The overall alignment was quantified by calculating the cosine similarity between the mean vectors

of the two corpora [26]. At the topic level, the top keywords from each topic were concatenated into a single text string, encoded using the same model, and compared using cosine similarity. To further explore the relationships between topics, hierarchical clustering was applied. Dendrograms and heatmaps were used to visualize these relationships, highlighting areas of convergence and divergence between academic and policy topics [27].

2.5. Structural coupling analysis

Structural coupling between academic and policy networks was evaluated to uncover the mechanisms of knowledge exchange. Keywords were extracted from each dataset using the RAKE algorithm, which demonstrated superior performance compared to traditional methods like TF-IDF [28]. Co-occurrence relationships among keywords were used to construct topic knowledge networks, where shared terms between academic and policy texts served as thematic bridges [29]. The similarity of node importance and edge weights within these networks was quantified using Hellinger distance [48], which provided measures of both node and edge coupling strength. The annual structural coupling strength was calculated as a weighted average of these metrics, offering temporal insights into the dynamics of knowledge exchange [9].

2.6. Dynamic decision analysis using deep reinforcement learning

Dynamic decision analysis was conducted using deep reinforcement learning (DRL) to prioritize themes in a dynamic and data-driven manner [30]. A custom environment, TopicSelectionEnv, was designed to simulate theme selection by integrating factors such as topic similarity, popularity, and structural metrics as inputs. The decision-making process was supported by a Deep Q-Network (DQN) implemented in PyTorch [31], featuring three fully connected layers to achieve a balance between computational efficiency and model complexity. The epsilon-greedy policy was employed during training to ensure a balance between exploration and exploitation. Iterative updates refined the Q-values based on discounted future rewards, enabling the model to prioritize themes with the highest Q-values, which represented optimal alignment with policy objectives.

3. Results

3.1. Alignment and interconnection between scientific and policy topics

3.1.1. Topic discovering

Viewed through the lens of the interplay between scientific development and policy evolution, groundbreaking advancements frequently originate from seemingly inconspicuous themes [9]. Consequently, beyond emphasizing progress in scientific inquiry, it is essential to focus on identifying meso- and micro-level themes that influence macro-level policy strategies [32]. In this context, the BERTopic model, is utilized to uncover pivotal themes in science and policy. The clustered results are summarized in Table 1. The first two columns illustrate the document counts associated with each identified topic, derived from the respective probability values. The third and fourth columns detail the percentage share of these topics among academic publications and their corresponding representation within policy texts. Lastly, the fifth and sixth columns provide the themes emerging from academic papers and policy documents, respectively.

The thematic analysis of research papers and policy documents unveils distinct yet overlapping priorities in the energy and environmental sectors. Topic 1, "Coal mining and geological formations," highlights the integration of geological and safety studies within academic research, which captures 15.73 % of the thematic distribution. This aligns with Topic 2, "Safety regulations and worker protection," representing 13.35 % of policy discussions, collectively emphasizing the shared commitment to mining safety and geological sustainability. At the intersection

Table 1

Clustered results and thematic distribution for academic publications and policy texts within each identified topic.

Topics	No. of documents		Ration (%)		Interpretations of the topics	
	Paper	Policy	Paper	Policy	Paper	Policy
1	12551	104	15.73 %	13.35 %	Coal mining and geological formations	Safety regulations and worker protection
2	6215	48	7.79 %	6.16 %	Surface dynamics and climate data	Renewable energy and grid management
3	4702	38	5.89 %	4.88 %	Building energy systems and performance	Energy conservation and efficiency
4	2426	33	3.04 %	4.24 %	Electric vehicles and transportation	Urban development and pollution policies
5	2364	30	2.96 %	3.85 %	Metabolism and microbial gene expression	Regional governance and energy coordination
6	2220	30	2.78 %	3.85 %	Carbon emissions and decoupling	Administrative responsibilities in energy management
7	1795	26	2.25 %	3.34 %	Modeling and forecasting techniques	Biomass and alternative energy resources
8	1743	26	2.18 %	3.34 %	Nuclear energy and neutron technology	Electricity usage and regulation
9	1605	24	2.01 %	3.08 %	Economic growth and energy consumption	Criminal penalties for electricity theft
10	1589	23	1.99 %	2.95 %	Power systems and renewable integration	Nuclear energy and low-carbon development
11	1395	22	1.75 %	2.82 %	Household energy consumption	Energy conservation practices and demand-side management
12	1289	20	1.62 %	2.57 %	Dietary intake and health metrics	Construction standards and investment approvals
13	1132	20	1.42 %	2.57 %	Polycyclic aromatic hydrocarbons	Industrial production and quality control
14	1018	18	1.28 %	2.31 %	Green innovation and environmental finance	National law and territory management
15	1013	17	1.27 %	2.18 %	Air pollution and public health	Energy-saving technologies and public policy
16	997	17	1.25 %	2.18 %	Particulate matter and source analysis	Technological innovation in energy development
17	929	17	1.16 %	2.18 %	Carbon tax and emission policies	Emergency response and compensation mechanisms
18	857	15	1.07 %	1.93 %	Energy security and policy cooperation	Inspection and administrative supervision
19	850	14	1.07 %	1.80 %	Water-energy nexus and resource use	Rural electrification and agricultural development
20	845	14	1.06 %	1.80 %	Energy efficiency and industrial performance	Construction operations and feasibility studies
21	812	13	1.02 %	1.67 %	Oil markets and price volatility	Sustainable energy development and security
22	715	12	0.90 %	1.54 %	Digital economy and carbon transformation	Regulatory violations and enforcement
23	605		0.76 %		Biomass and bioenergy production	
24	584		0.73 %		Heavy metals and soil pollution	
25	568		0.71 %		Risk evaluation and decision methods	
26	426		0.53 %		Renewable power markets and costs	
27	402		0.50 %		Municipal solid waste management	

of climate adaptation and energy optimization, Topic 3, “Surface dynamics and climate data” (7.79 %), and Topic 4, “Renewable energy and grid management” (6.16 %), underline efforts to develop data-driven solutions for energy systems while addressing climate challenges.

Energy efficiency and emissions reduction emerge as central themes across both domains. Topic 5, “Building energy systems and performance” (5.89 %), underscores academic focus on optimizing energy utilization, while Topic 6, “Energy conservation and efficiency” (4.88 %), reflects parallel policy measures to mitigate energy waste. The shared challenge of balancing economic growth and environmental sustainability is encapsulated in Topic 7, “Carbon emissions and decoupling” (2.78 % in academic studies and 3.85 % in policy discussions). These efforts are supported by technological innovations, as seen in Topic 8, “Modeling and forecasting techniques” (2.25 %), and Topic 9, “Biomass and alternative energy resources” (3.34 %), which collectively advance renewable energy solutions and predictive analytics.

The socio-economic dimensions of energy governance are also evident. Topic 10, “Economic growth and energy consumption” (2.01 %), explores the intricate relationship between energy utilization and economic development, while Topic 11, “Criminal penalties for electricity theft” (3.08 %), addresses governance challenges within energy systems. Moreover, Topic 12, “Green innovation and environmental finance” (1.28 %), and Topic 13, “National law and territory management” (2.31 %), reflect a growing focus on sustainability-oriented innovation and regulatory frameworks.

Emerging topics signal areas for further exploration. For instance, Topic 14, “Digital economy and carbon transformation” (0.90 %), hints at the nascent application of digital solutions to decarbonization. Similarly, niche topics like Topic 15, “Biomass and bioenergy production” (0.76 %), and Topic 16, “Heavy metals and soil pollution” (0.73 %), indicate the need for deeper research into specific environmental challenges.

Sector-specific insights further enrich this analysis. Policy-oriented themes such as Topic 17, “Urban development and pollution policies” (4.24 %), and Topic 18, “Rural electrification and agricultural development” (1.80 %), reveal strategic energy governance approaches tailored to regional needs. These are complemented by academic

explorations of foundational areas, including Topic 19, “Metabolism and microbial gene expression” (2.96 %), and Topic 20, “Energy security and policy cooperation” (1.07 %).

3.1.2. Topic popularity of the scientific papers

This analysis adopts a holistic methodology to assess topic popularity through the application of Equations (1)–(4). As the time series data is solely available within the paper dataset, the computation of topic popularity is limited to this corpus. The findings are presented in Fig. 2.

The analysis of topic popularity reveals clear patterns, with some topics showing dominant attention and others reflecting emerging or niche interests. Topic 1 is the most prominent, achieving the highest popularity score ($P^k = 1.34$). This is primarily due to its strong temporal trend ($S_{NT}^k = 1$), indicating sustained and growing interest, along with a notable normalized proportion ($S_{NP}^k = 0.34$). Similarly, Topic 5 ($P^k = 0.81$) benefits from a strong temporal trend ($S_{NT}^k = 0.65$), even though its proportion score ($S_{NP}^k = 0.16$) is lower. Topic 2, with a balanced contribution from both metrics ($S_{NT}^k = 0.47$), also ranks high ($P^k = 0.88$).

Moderately popular topics reflect varying levels of focus. For example, Topic 3 ($P^k = 0.69$) shows a stronger normalized proportion ($S_{NP}^k = 0.45$) but a lower temporal trend ($S_{NT}^k = 0.24$). In contrast, Topic 13 ($P^k = 0.54$) relies more on its proportion score ($S_{NP}^k = 0.38$) than its temporal trend ($S_{NT}^k = 0.16$). Similarly, Topic 12 ($P^k = 0.44$) gains moderate attention due to a noticeable temporal trend ($S_{NT}^k = 0.34$), despite a smaller proportional contribution ($S_{NP}^k = 0.11$). At the other end of the spectrum, some topics show limited attention. Topic 22 has the lowest popularity score ($P^k = 0.13$), with a proportion score of 0.13 and no observable temporal trend ($S_{NT}^k = 0$). Similarly, Topic 10 ($P^k = 0.17$) and Topic 26 ($P^k = 0.16$) show minimal engagement, as both metrics contribute weakly. Emerging topics suggest areas for future exploration. For instance, Topic 27 ($P^k = 0.43$) has a relatively high proportion score ($S_{NP}^k = 0.36$) but a lower temporal trend ($S_{NT}^k = 0.07$). Topic 18 ($P^k = 0.37$) and Topic 24 ($P^k = 0.37$) also show moderate scores, reflecting balanced contributions from both metrics.

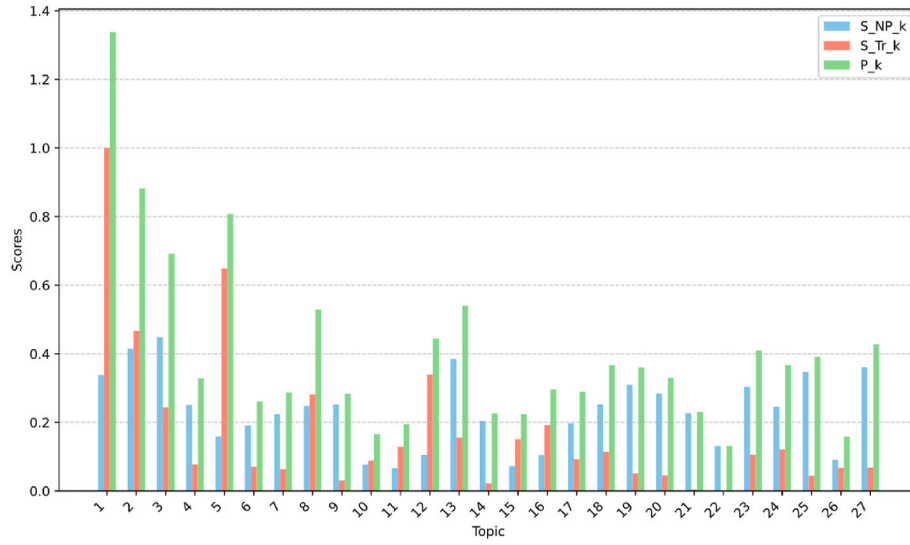


Fig. 2. A comprehensive assessment of topic popularity in published papers. The popularity score P^k is determined by integrating two factors: the normalized proportion score S_{NP}^k and the normalized temporal trend score S_{NTr}^k .

Temporal trends offer additional insights. Some topics, such as Topic 4 ($P^k = 0.33$) and Topic 9 ($P^k = 0.28$), have relatively high proportions ($S_{NP}^k = 0.25$) but low temporal trends ($S_{NTr}^k = 0.08$ and $S_{NTr}^k = 0.03$, respectively), indicating static or declining interest over time. On the other hand, topics like Topic 6 ($P^k = 0.26$) and Topic 7 ($P^k = 0.29$) maintain modest temporal trends ($S_{NTr}^k = 0.07$ and $S_{NTr}^k = 0.06$) despite lower proportions, suggesting potential for future growth. In summary,

the findings demonstrate a dynamic thematic landscape. Dominant topics reflect sustained interest; while emerging and niche topics highlight areas of potential growth.

3.1.3. Semantic similarity calculations

After obtaining the respective topic distributions and popularity for both policy and scientific papers in the context of semantic overlap, a more comprehensive analysis was conducted by comparing the semantic

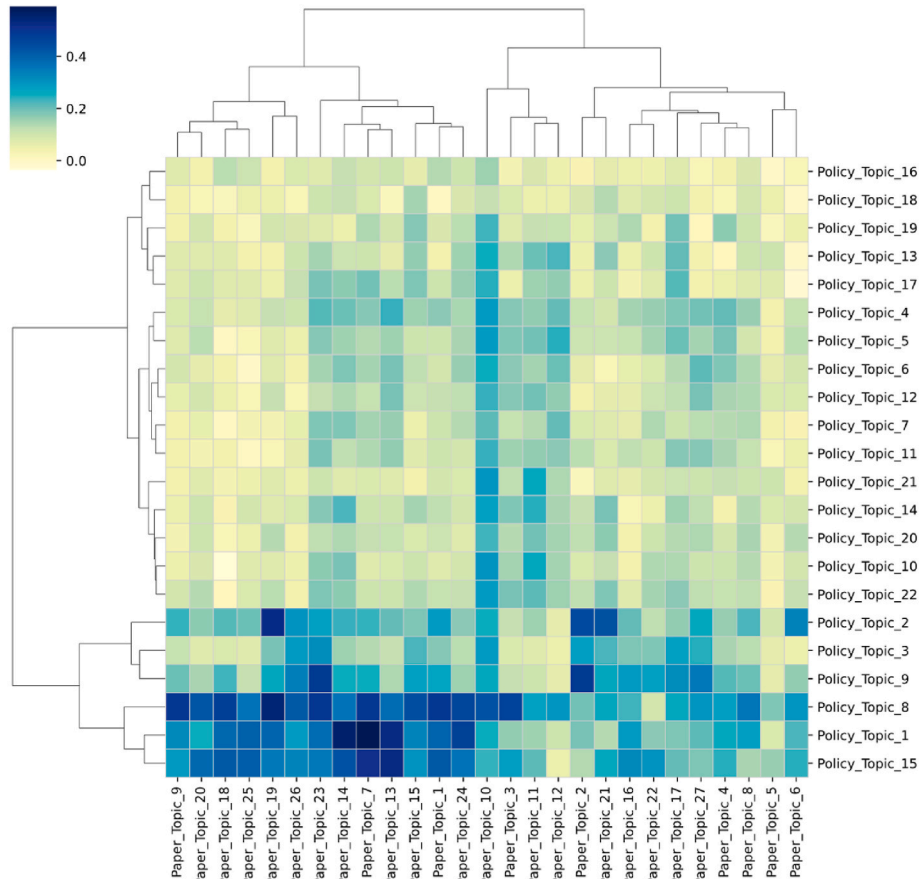


Fig. 3. Topic similarity and between scientific papers and policy documents.

similarity of the entire text and the thematic-level similarity. A heatmap of the cosine similarity matrix (Fig. 3) visually illustrates these findings. High-similarity clusters are evident, with darker regions indicating strong alignment between specific policy and paper topics. Conversely, lighter regions highlight areas of weak or negative alignment. The clustering of highly related topics suggests concentrated areas of thematic overlap, whereas the dispersion of low-similarity pairs reveals potential gaps in academic and policy integration. The analysis quantifies the thematic alignment between 22 policy topics and 27 paper topics, providing insights into areas of convergence and divergence. Below is a systematic summary of the results.

From the perspective of hierarchical clustering, as shown in Fig. 3, *policy_topic_2*, *policy_topic_3*, and *policy_topic_9* are grouped together, indicating a very high similarity among these topics. Similarly, *policy_topic_1*, *policy_topic_8*, and *policy_topic_15* are also clustered, suggesting a strong thematic alignment between these topics. Notably, *paper_topic_10* demonstrates a moderate degree of relevance to the majority of the policy topics, which is an observation worthy of attention.

From the perspective of topic similarity, the analysis quantifies the thematic alignment between 22 policy topics and 27 paper topics, offering insights into areas of convergence and divergence. The average cosine similarity across all topic pairs was 0.23 (SD = 0.15), with values ranging from −0.035 to 0.59. This moderate overall alignment indicates that while certain topics exhibit strong thematic overlap, others display significant divergence. Notably, 12.5 % of all topic pairs showed high similarity (cosine similarity >0.4), reflecting well-aligned themes, while 8.2 % exhibited low or negative similarity (cosine similarity <0.1), highlighting areas of thematic disconnect.

Among the top 10 most highly related topic pairs, the strongest alignment was observed between *Safety regulations and worker protection (policy_topic_1)* and *nuclear energy and neutron technology (paper_topic_7)*, with a cosine similarity of 0.59. This pairing underscores a shared emphasis on nuclear safety and worker protection, suggesting robust thematic convergence in these domains. Similarly, *safety regulations and worker protection (policy_topic_1)* and *air pollution and public health (paper_topic_14)* showed a similarity of 0.56, highlighting a close connection between occupational safety and environmental health in both policy and research.

Another high-similarity pair, *electricity usage and regulation (policy_topic_8)* and *energy security and policy cooperation (paper_topic_19)*, achieved a similarity of 0.55, demonstrating the alignment of industrial energy regulation policies with academic research on energy security. The overlap between *safety regulations and worker protection (policy_topic_1)* and *green innovation and environmental finance (paper_topic_13)*, with a similarity of 0.53, reflects the integration of safety regulations with innovative financing strategies for environmental sustainability. Similarly, *energy-saving technologies and public policy (policy_topic_15)* and *green innovation and environmental finance (paper_topic_13)*, also with a similarity of 0.53, underscore the alignment of public policies promoting energy-saving technologies with research on green innovation. Other high-similarity pairs revealed strong thematic connections in areas such as *sustainable energy development*, *economic growth*, and *renewable integration*, indicating concentrated areas of policy-research alignment that could foster collaborative opportunities.

In contrast, the least related topic pairs revealed significant thematic divergence. The pair *nuclear energy and low-carbon development (policy_topic_10)* and *water-energy nexus and resource use (paper_topic_18)* exhibited the lowest similarity, with a cosine value of −0.035, suggesting a disconnect between nuclear energy policies and research on the water-energy nexus. Similarly, *emergency response and compensation mechanisms (policy_topic_17)* and *modeling and forecasting techniques (paper_topic_6)*, with a similarity of −0.028, highlighted the limited application of modeling techniques to emergency response policies. Lastly, *industrial production and quality control (policy_topic_13)* and *modeling and forecasting techniques (paper_topic_6)*, with a similarity of −0.009, demonstrated minimal overlap, reflecting a lack of connection

between industrial quality control policies and forecasting methodologies.

3.2. Structural coupling of the policy and scientific knowledge networks

Utilizing RAKE, a total of 413,956 keywords were extracted from published papers and 618 keywords from policy documents. However, simply comparing keywords between the two databases is far from sufficient. Following the multi-perspective analysis of themes in Section 3.1, this section establishes another type of knowledge network. By comparing coupling results, it aims to validate or supplement the findings of the previous method. Table 2 presents the 10 most common keywords, highlighting the differing priorities between science and policy in the energy field in China.

The keywords used in policy documents are more governance-oriented, focusing on terms such as state council, public participation, and legal liability, which emphasize administrative responsibilities, societal engagement, and regulatory enforcement. These terms reflect the practical and operational focus of policymaking, addressing immediate challenges and implementing actionable solutions. In contrast, academic papers use more problem-driven terms, such as energy efficiency, climate change, and sustainable development, which are geared toward exploring underlying issues and proposing innovative solutions from a broader, analytical perspective. This divergence highlights how the two domains approach the energy field: policy documents prioritize actionable governance, while academic research seeks to understand and address systemic issues from a macro perspective.

After obtaining the keyword distribution, a temporal knowledge network was created to analyze the structural coupling between the policy and scientific networks. This study organized academic papers and policy documents by year, constructing corresponding knowledge networks for the past 20 years. Since the policy network lacks a clear temporal sequence, its size values may exhibit similarities across different years. Table 3 provides the descriptive statistics for these networks.

The comparative analysis highlights key differences between scientific and policy networks, as summarized in Table 3. Scientific networks are significantly larger, with an average of 19,867 nodes (ranging from 2114 to 56,537) and 64,646 edges (ranging from 5053 to 205,094). In contrast, policy networks are smaller, with 591 nodes and 2029 edges, showing minimal variation across years. This reflects the broader and more diverse focus of scientific networks compared to the targeted and consistent themes in policy networks.

Policy networks exhibit much higher density (0.011 vs. 0.00069) and clustering coefficients (0.12 vs. 0.004), indicating stronger internal cohesion and tightly connected structures. Additionally, policy networks have significantly fewer components (8.85, ranging from 3 to 14) compared to the highly fragmented scientific networks, which average

Table 2

The 10 most commonly occurring keywords identified in both academic papers and policy documents by RAKE.

Rank	Policy	Frequency	Papers	Frequency
1	environmental protection	92	energy consumption	6868
2	state council	42	economic growth	4587
3	environmental pollution	34	sustainable development	3404
4	environmental protection authority	22	renewable energy	3233
5	public participation	17	energy efficiency	3182
6	ecological damage	14	economic development	2912
7	county level	13	climate change	2902
8	information disclosure	11	case study	2462
9	legal liability	10	coal combustion	2230
10	social development	9	natural gas	1954

Table 3

Summary statistics for internal network characteristics.

Network Properties	Scientific networks			Policy networks		
	Mean	Min	Max	Mean	Min	Max
Nodes	19867	2114	56537	591	591	591
Edges	64646	5053	205094	2029	2011	2060
Density	6.9E-04	1.2E-04	22E-04	1.1E-02	1.1E-02	1.1E-02
Avg Clustering Coefficient	0.004	0.002	0.01	0.12	0.10	0.13
Components	295.7	39	822	8.85	3	14
Largest Component Nodes	1956	21	9772	120	118	122
Largest Component Edges	19567.3	2072	55705	583.1	578	589
Diameter	10.8	10	12	6.45	6	8
Largest Component Clustering Coefficient	0.0046	0.0021	0.013	0.123	0.108	0.133
Avg Shortest Path Length	4.51	3.94	4.85	2.97	2.93	3.02

295.7 components (ranging from 39 to 822). The largest components in policy networks include 120 nodes and 583 edges, with a clustering coefficient of 0.123, while the largest components in scientific networks, though much larger (1956 nodes and 19,567 edges), exhibit a clustering coefficient of only 0.0046.

The average shortest path length further highlights the structural differences, with policy networks achieving a shorter path (2.97) compared to scientific networks (4.51). These results indicate that policy networks enable more efficient knowledge transfer and communication, likely due to their smaller size and focus on actionable themes.

These findings demonstrate the complementary roles of the two network types: scientific networks provide a foundation for exploring diverse topics, while policy networks focus on consolidating knowledge into cohesive frameworks for decision-making. Aligning the broad insights from scientific networks with the targeted focus of policy networks can foster evidence-based policies and effective knowledge integration.

3.3. Deep reinforcement learning decision analysis

Through comprehensive analyses detailed in Sections 3.1 to 3.4, this study progressively explored the topic distributions within the two corpora, examined the prominence of paper topics, evaluated the topic similarities between the two datasets, and mapped the semantic structures of the topics. In the concluding phase, these findings were incorporated into a deep reinforcement learning framework to conduct decision-making analysis, with the results illustrated in Fig. 4.

The analysis evaluated the performance of 49 themes in the Paper Corpus using Q-values as the primary metric to assess optimization effectiveness. The results identified Theme 31 *power systems and renewable integration* as the optimal choice, achieving the highest Q-value of 1.6225, indicating its superior alignment with the objective of the study. Other high-performing themes included Theme 5 *regional governance and energy coordination* (1.5768), Theme 46 *heavy metals and soil pollution* (1.3920), and Theme 3 *energy conservation and efficiency*

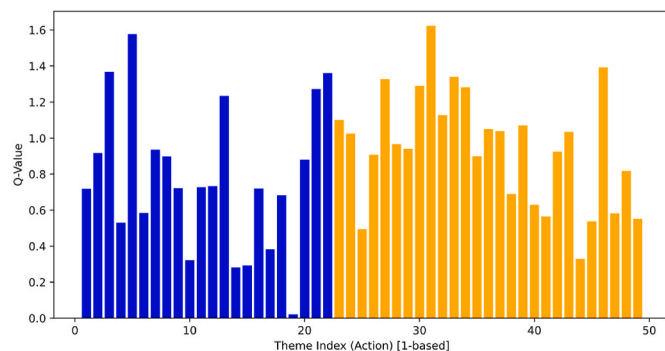


Fig. 4. Deep reinforcement learning results. Blue collum represents policy corpus. Orange represents paper corpus. Q-value for all 22 topics.

(1.3682), which demonstrated competitive yet slightly lower optimization scores. These themes stand out as potential areas of focus for further exploration or implementation.

In contrast, some themes exhibited significantly lower Q-values, such as Theme 19 *rural electrification and agricultural development* (0.0211), Theme 14 *national law and territory management* (0.2813), and Theme 10 *nuclear energy and low-carbon development* (0.3219). These themes reflect minimal contribution to the optimization goal and may require further refinement or re-evaluation to enhance their relevance. The overall range of Q-values spanned from 0.0211 to 1.6225, with a substantial concentration of themes scoring between 0.5 and 1.0. Notably, approximately 20 % of themes achieved Q-values exceeding 1.0, highlighting a subset of promising themes with strong performance.

4. Discussion

This study offers a comprehensive analysis of thematic alignment and structural dynamics between academic research and policy-making in the Chinese energy sector. By employing advanced methodologies, including BERTopic and deep reinforcement learning (DRL), it provides actionable insights into the interplay between these domains. For instance, the prominence of “Coal mining and geological formations” in both academic research (15.73 %) and policy discourse (13.35 %) underscores its strategic importance. Academic research emphasizes geological sustainability and technological innovations, while policy frameworks prioritize immediate safety measures and worker protection. Bridging this gap presents an opportunity to integrate predictive safety models derived from academic studies into policy-making, potentially enhancing both operational efficiency and safety outcomes. Such integration aligns with Yu et al. [9], who underscore the necessity of interdisciplinary approaches for addressing complex energy governance challenges.

Similarly, “Energy conservation and efficiency” highlights complementary but divergent priorities. While academic research pioneers advanced energy optimization technologies, policy-making focuses on conservation measures and demand-side management [33]. To address this misalignment, targeted subsidies for energy-efficient technologies and tax incentives for retrofitting older buildings could play a crucial role in bridging the gap between academic research and practical implementation. For instance, financial incentives can accelerate the adoption of advanced energy optimization technologies developed through academic research [34], while tax policies can encourage the application of these innovations in upgrading older infrastructure [35]. Germany’s experience with policy-driven energy efficiency programs demonstrates how combining financial incentives with regulatory standards has successfully reduced energy consumption in residential and industrial sectors [36]. Such examples provide a roadmap for China to align academic advances with its energy efficiency goals.

“Nuclear energy and neutron technology” illustrates a moderate alignment, with policies emphasizing safety and energy regulation while academic research explores advanced neutron applications.

Strengthening collaboration through joint funding initiatives and pilot projects could close this gap. It is also important to note that the moderate alignment observed in this theme is partially shaped by China's unique policy framing. Unlike many Western countries where nuclear and renewable energy are considered separate domains, Chinese energy policy often groups nuclear power together with renewables under the broader classification of low-carbon or alternative energy sources [37]. This institutional framing influences both policy narratives and academic discourse, resulting in a semantic overlap between nuclear-related themes and broader renewable integration strategies. In this study, nuclear energy was analyzed within the same analytical category as renewables to maintain consistency with domestic policy structures. This approach ensures semantic alignment within the Chinese context but may introduce challenges when comparing alignment metrics across countries with differing energy taxonomies.

The structural analysis of knowledge networks reveals distinct yet complementary characteristics. Scientific networks, while expansive and diverse, often lack the coherence needed for direct policy application, whereas policy networks, characterized by higher density (0.011) and clustering coefficients (0.12), excel in efficient knowledge dissemination but may struggle with adaptability to emerging challenges. For example, UK and Germany's distributed energy system policies showcase how tightly integrated networks can enable rapid decision-making while maintaining flexibility to incorporate innovative technologies [38]. Establishing collaborative energy platforms, such as cross-sector working groups or integrated research-policy hubs, could align the exploratory nature of scientific networks with the targeted efficiency of policy networks, enhancing evidence-based policy-making [49].

Temporal dynamics highlight the sustained importance of established themes, such as "Coal mining and geological formations" ($P^k = 1.34$), alongside the emergence of transformative areas like "Digital economy and carbon transformation" ($P^k = 0.13$). The latter signals opportunities to leverage digital technologies for optimizing carbon reduction strategies [39]. For instance, digital twins could simulate energy usage scenarios in industrial processes, enabling real-time adjustments to minimize emissions [40]. Policies incentivizing the adoption of these technologies through grants or tax breaks could place China at the forefront of global energy transitions. Song et al. [41] similarly emphasize the importance of aligning policies with emerging trends to remain responsive to evolving thematic landscapes.

Semantic similarity analysis further uncovers areas of alignment and divergence between academic research and policy-making [42]. High-similarity themes, such as "Safety regulations and worker protection" and "Nuclear energy and neutron technology," reflect opportunities for immediate collaboration. Conversely, low-similarity pairs, such as "Nuclear energy and low-carbon development" and "Water-energy nexus and resource use," highlight thematic gaps that warrant targeted interventions. Bridging these gaps could involve interdisciplinary collaborations through think-tank-led policy dialogues and co-funded research initiatives [43]. For example, India's success in addressing rural electrification challenges through public-private partnerships demonstrates how cross-sector collaboration can effectively align academic insights with policy priorities [44].

Deep reinforcement learning (DRL), which has been proven effective across various domains [45], provides a novel approach in this analysis, facilitating the dynamic prioritization of themes based on real-time feedback. This analysis identified "Power systems and renewable integration" as the most relevant theme (Q-value = 1.6225). Practical implementation could include financial incentives for distributed renewable systems, investments in grid modernization, and support for large-scale energy storage solutions. Conversely, themes like "Rural electrification and agricultural development" demonstrate limited optimization potential under current conditions. However, lessons from some developing countries' electrification initiatives reveal that targeted investments in rural energy infrastructure, combined with

adaptive policy frameworks, can transform these themes into critical drivers of socioeconomic development [46]. This adaptability underscores DRL's potential as a flexible tool for addressing complex policy challenges [47].

Beyond its immediate contributions, this study establishes a scalable framework for aligning academic research with policy-making priorities. By integrating BERTopic, semantic similarity analysis, and DRL, it offers a robust method for identifying actionable pathways, addressing thematic gaps, and anticipating emerging challenges. The framework's adaptability also allows for application in other policy areas, such as environmental governance and social equity, where similar techniques could guide biodiversity conservation or climate adaptation strategies.

Although the framework involves a certain degree of technical complexity, it is intended to serve as a flexible decision-support tool to assist policymakers in prioritizing topics based on science-policy alignment. Its outputs—such as ranked alignment scores or thematic gap maps—could be translated into simplified formats like dashboards or policy briefs to enhance usability. When applying this approach to other domains, such as environmental governance or social equity, challenges such as sector-specific data availability, stakeholder diversity, and regulatory complexity may arise. Future research could explore tailored adaptation strategies to ensure that the framework remains both analytically rigorous and practically useable across diverse policy contexts, thereby supporting more responsive and evidence-informed decision-making.

5. Conclusion and limitations

Integrating large-scale text mining with network analytics and deep reinforcement learning, the framework quantitatively maps the interplay between 106,661 English-language research articles and 618 national-level energy policies, revealing a landscape in which thematic convergence coexists with critical blind spots. Dominant overlaps—such as the shared focus on "Coal mining and geological formations" and the jointly prioritised "Safety regulations and worker protection"—contrast sharply with nascent yet strategic domains like "Digital economy and carbon transformation," whose popularity score is a mere 0.13 and whose academic-policy cosine similarity rarely exceeds 0.10. Structural diagnostics show policy networks to be 15.9 times denser and to exhibit clustering coefficients 30-fold higher than their scientific counterparts, underscoring a governance architecture that is cohesive but thematically narrow. Deep Q-learning further ranks "Power systems and renewable integration" as the most policy-actionable theme ($Q = 1.6225$), while signalling low optimization returns for "Rural electrification and agricultural development," thereby providing an evidence-based hierarchy for resource allocation. Together, these findings furnish a scalable, data-driven blueprint for synchronising research investment and regulatory agendas, while highlighting the need to incorporate multilingual corpora, regional policies and longitudinal dynamics to capture the full complexity of China's energy transition and inform responsive governance in other high-stakes domains.

Despite its contributions, this study has several limitations. It relies solely on English-language academic publications from international databases (e.g., Web of Science) and centralized national-level policy documents. While this ensures data quality and consistency, it excludes Chinese-language research, regional policies, and grey literature, potentially overlooking grassroots initiatives and local governance dynamics. This may limit the generalizability and contextual depth of the findings. Future studies should incorporate multilingual and decentralized data sources to enhance comprehensiveness. Additionally, treating policies as static entities fails to capture their temporal evolution. Longitudinal analyses could offer deeper insights into the co-development of research and policy. Although the proposed DRL framework shows promise for dynamic prioritization, its complexity may hinder adoption. Simplifying its structure and improving usability could support broader policy application. Lastly, the energy sector focus limits generalizability;

extending this framework to fields like public health or urban planning would help validate its cross-domain utility. Incorporating expert validation or hybrid modeling could also enhance the precision of thematic interpretations.

Declaration of generative AI in scientific writing

After completing the writing of this manuscript, the author used ChatGPT-4 solely to improve the language and enhance readability. The tool was not used to generate original content, and the author carefully reviewed and edited the output as necessary. The author takes full responsibility for the content of the published article.

Declaration of competing interest

No conflict of interest statement.

Data availability

Data will be made available on request.

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